

C6 Project 2

Ball action Spotting

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MASTER IN
COMPUTER VISION
Barcelona



GitHub Repository

<https://github.com/mcv-m6-video/mcv-c6-2026-team3>

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1. Introduction

Tasks objective

Action classification (W5)

Multi-label classification of football actions
per clip

Action spotting (W6 & W7)

Per-frame classification of football actions
(only one per frame)

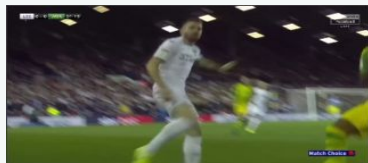
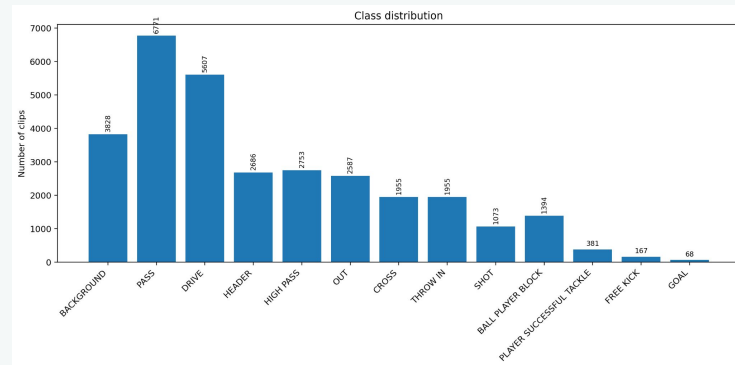
Dataset - SoccerNet

- 7 football matches at 720p (25 fps)
- 4 training, 1 validation and 2 test videos
- 12 action labels for the action spotting task

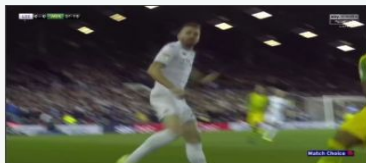


2. Dataset challenges

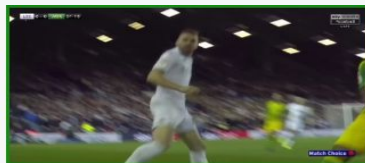
- Class Imbalance
- Not homogeneous frames
- Difficult GT



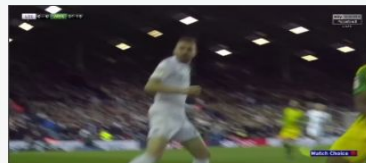
BALL PLAYER BLOCK
train | GT frame 1957 | 1 - 01:18



t-1 | f1956



t | f1957



t+1 | f1958



t+2 | f1959

Not Homogenous frames: Mostly from above, few are near to the players



BALL PLAYER BLOCK
train | GT frame 615 | 1 - 00:24



t-1 | f614



t | f615



t+1 | f616



t+2 | f617

Difficult GT: Where is the ball?

3. Task challenges

Action-specific challenges

- Distinguish between visually similar actions
- Recognition from multiple viewpoints
- Co-occurring actions
- Crowded background

Action Spotting challenges

- Anchor point ambiguity (when does an action occur?)
- Extremely sparse labeling (most frames are background)
- Classification temporal tolerance (how far from the GT is a prediction still correct?)

4. Background work (W5 & W6)

Week 5 - Take Home Messages

Default to Native 3D Architectures

2D models with temporal pooling fail on sustained actions (like throw-ins). For multi frame classification, spatio temporal backbones are much better.

Use Clip-Consistent Augmentations

To the image consistency found on videos, any spatial data augmentation must be applied identically to every single frame within a given clip.

Standard Class Rebalancing Fails Here

When dealing with a really high class imbalance, traditional fixes like weighted loss or stratified sampling backfire and degrade performance.

Enormous Compute Trade-offs

The high accuracy brought by 3D models come in exchange for enormous increases on the resources they need, needing 10x more parameters and 50x more GMACs compared to the baseline.

Baseline

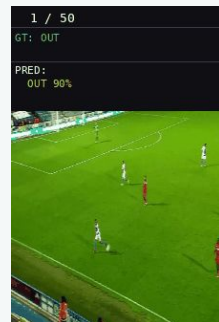
AP10= 31.15, AP12 = 27.25



Class	Average Precision
PASS	76.47
DRIVE	74.59
HEADER	22.85
HIGH PASS	56.29
OUT	13.52
CROSS	28.38
THROW IN	18.62
SHOT	17.17
BALL PLAYER BLOCK	6.07
PLAYER SUCCESSFUL TACKLE	1.58
FREE KICK	10.7
GOAL	0.82

Best Model: r2plus1d_18

AP10= 44.98, AP12 = 41.31



Class	Average Precision
PASS	86.94
DRIVE	77.98
HEADER	42.83
HIGH PASS	58.51
OUT	22.4
CROSS	52.71
THROW IN	43.88
SHOT	40.59
BALL PLAYER BLOCK	11.91
PLAYER SUCCESSFUL TACKLE	2.06
FREE KICK	33.33
GOAL	12.56

5. Background work (W5 & W6)

Week 6 - Take Home Messages

Validate on Your Target Metric

Validation loss is a poor proxy for spotting performance. Always save your best model epoch based on the actual metric you are optimizing for (in this case, AP10).

Not all can be transferred

Don't blindly copy classification tricks. Clip-wise augmentations actively harmed the spotting model's performance opposed to their behaviour on the classification task.

Capture Transitions with Feature Deltas

Spotting is about pinpointing state changes in a video. Calculating and concatenating consecutive frame differences is an efficient way to force the model to look at motion rather than static states.

Visual "Cheats"

Spotting models will use shortcuts—like predicting a "DRIVE" simply because a player is standing near the ball. Future models need deeper temporal context to learn actual motion instead of just pixel arrangements.

Baseline

AP10= 7.49, AP12 = 6.24



Class	Average Precision
PASS	29.36
DRIVE	17.45
HEADER	5.07
HIGH PASS	12.01
OUT	1.3
CROSS	1.86
THROW IN	4.17
SHOT	2.87
BALL PLAYER BLOCK	0.78
PLAYER SUCCESSFUL TACKLE	0
FREE KICK	0
GOAL	0

Best Model:

RegNetY-004 + TCN + Delta

AP10= 36.02, AP12 = 35.82



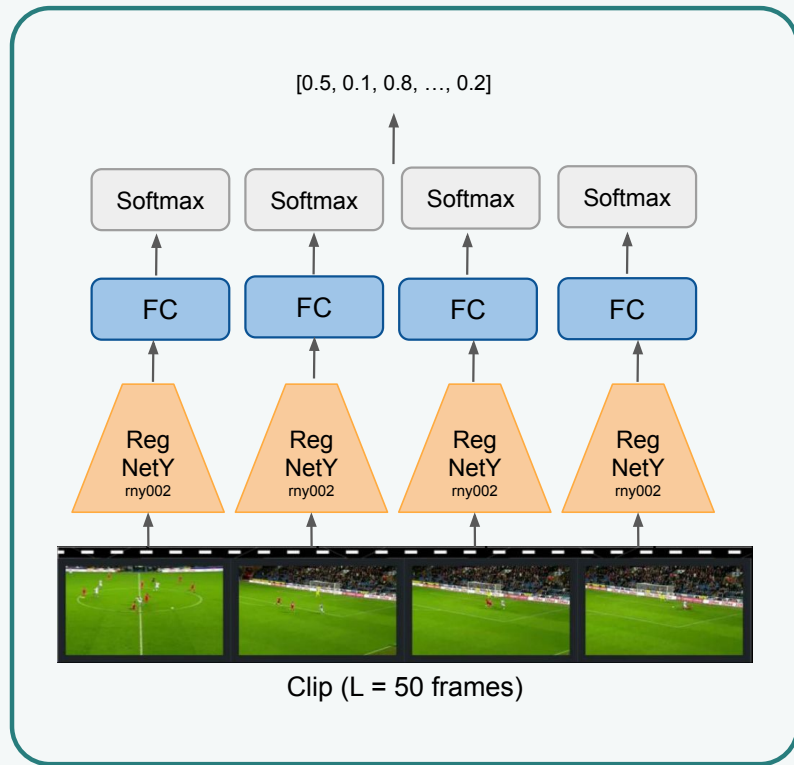
Class	Average Precision
PASS	71.16
DRIVE	66.05
HEADER	46.53
HIGH PASS	63.37
OUT	9.35
CROSS	41.1
THROW IN	24.31
SHOT	25.14
BALL PLAYER BLOCK	13.18
PLAYER SUCCESSFUL TACKLE	0
FREE KICK	69.7
GOAL	0

6. Ablation: Baseline

INSIGHTS

- Tested original model
- Low overall results
- Leverage of temporal information clearly needed

Variant	AP10	AP12
Baseline	7.49	6.24

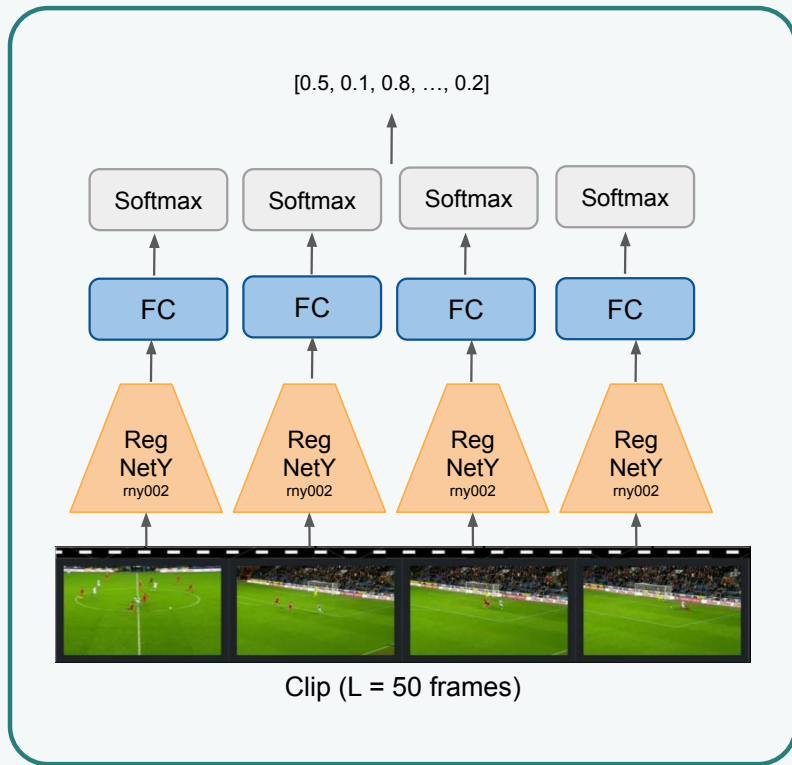


6. Ablation: AP as validation metric

INSIGHTS

- Misalignment: Validation Loss & AP
- Shift: Checkpoint selection using AP
- Impact: AP10 from 7.49 to 9.78

Variant	AP10	AP12
Baseline	7.49	6.24
+ AP	9.78	8.15

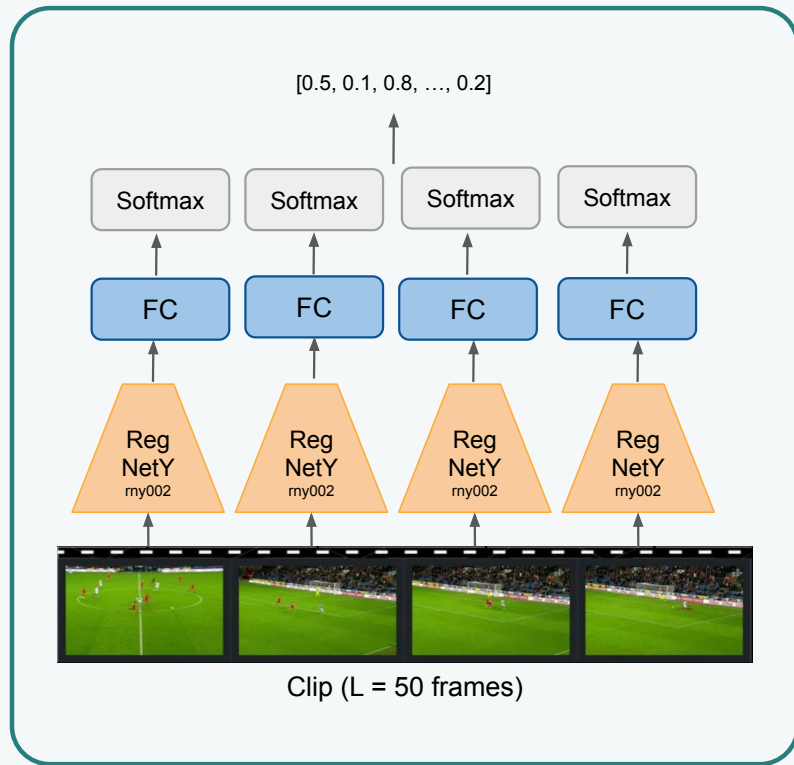


6. Ablation: Clip-level augmentation fail

INSIGHTS

- Changed augmentation approach to clip-level
- Results did not improve
- Kept original augmentation strategy

Variant	AP10	AP12
Baseline + AP	9.78	8.15
+ AUG	9.22	7.68

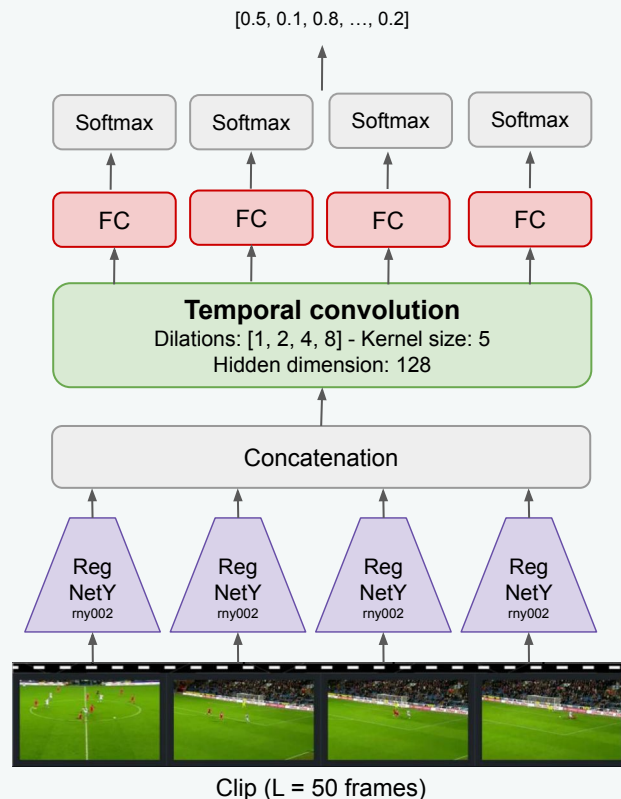


6. Ablation: Temporal convolution

INSIGHTS

- Added temporal convolutions
- Big jump on performance
- Leveraging temporal information increases performance massively.

Variant	AP10	AP12
Baseline + AP	9.78	8.15
+ Temporal	28.03	27.90

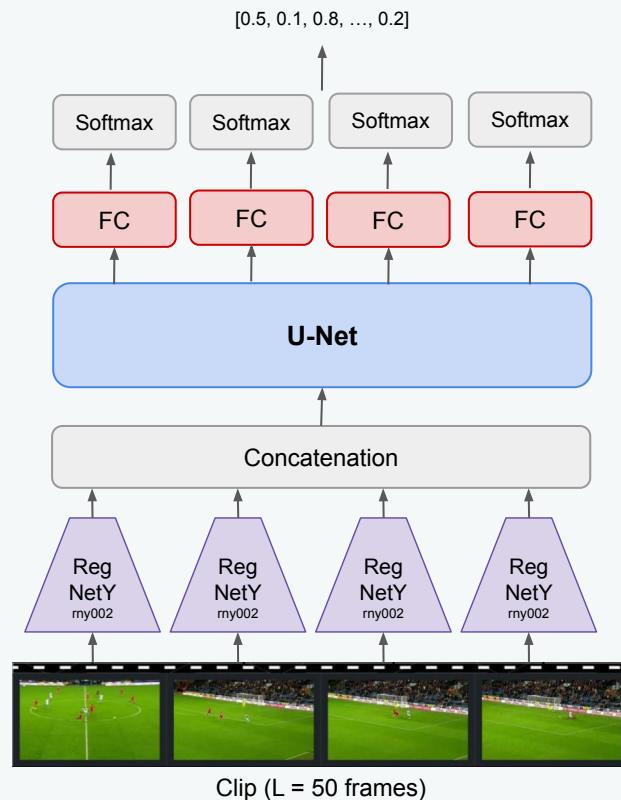


6. Ablation: U-Net - Max pool, Stride and Depth

INSIGHTS

- Max pooling is worse for lowering temporal resolution
- Added depth only improves slightly
- Smaller improvement than TCN and more parameters needed (3.7M vs 4.4M)

Variant	AP10	AP12
Max pool (depth=2)	20.69	17.24
Max pool (depth=3)	20.90	17.42
Strided (depth=2)	21.47	17.89
Strided (depth=3)	22.57	18.81



6. Ablation: Backbone

INSIGHTS

- Tested different backbone sizes
- Increasing the backbone capacity leads to better results for both approaches
- Increasing the backbone capacity too much leads to diminishing results, probably because overfitting.

Variant	AP10	AP12
Baseline + Temporal	28.03	27.90
+ rynn004 backbone	36.02	35.82
+ rynn008 backbone	34.14	35.52

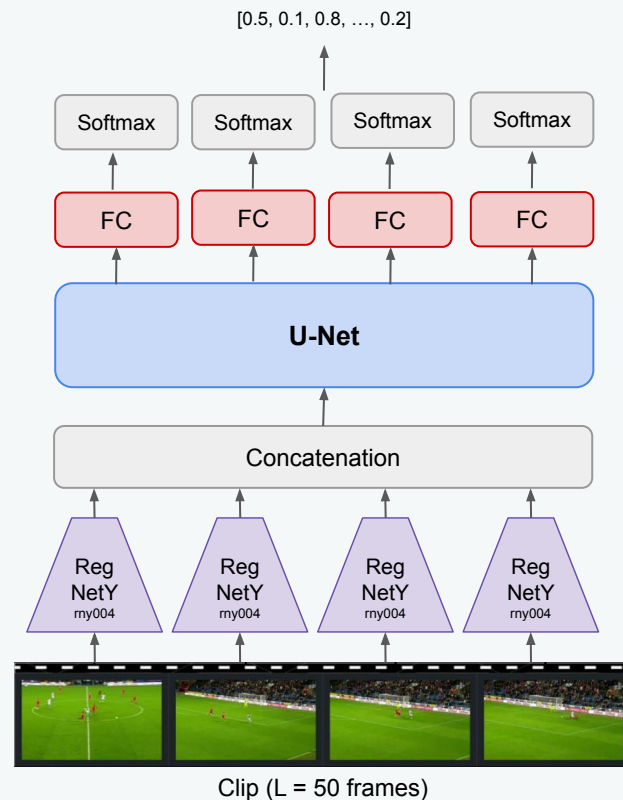
Variant	AP10	AP12
Baseline + U-Net	22.57	18.81
+ rynn004 backbone	31.22	26.01

6. Ablation: Soft labels

INSIGHTS

- Tested different distributions
- Simple softening works better than gaussian distribution

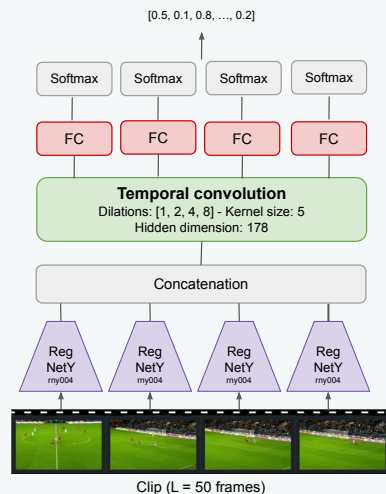
Variant	AP10	AP12
Baseline + U-Net + rynn004	31.22	26.01
+ Simple soft label	29.18	25.83
+ Gaussian soft label	28.78	25.50



7. Temporal aggregation vs MAP tolerances

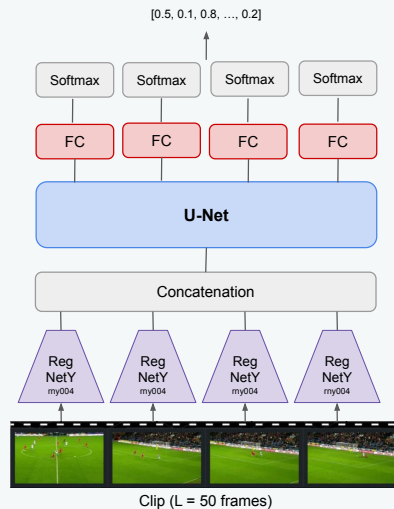
INSIGHTS

- Gap between tolerances a bit lower for the TCN model
- Similar temporal localization capacity (relative to their performances)
- Temporal convolution model superior compared to U-Net



Tolerance	AP10	AP12
Coarse (1s)	35.80	36.90
Tight (0.5s)	31.09	32.98

Parameters: 5.57M



Tolerance	AP10	AP12
Coarse (1s)	31.22	26.01
Tight (0.5s)	27.23	22.70

Parameters: 5.52M

8. Best action spotting model

The Winning Combination: TCN + Feature Deltas

Dual-Stream Context

By concatenating spatial features with temporal deltas, the model simultaneously processes state (player/ball position) and motion (transitions) before feeding it to the temporal head.

Preserved Temporal Resolution

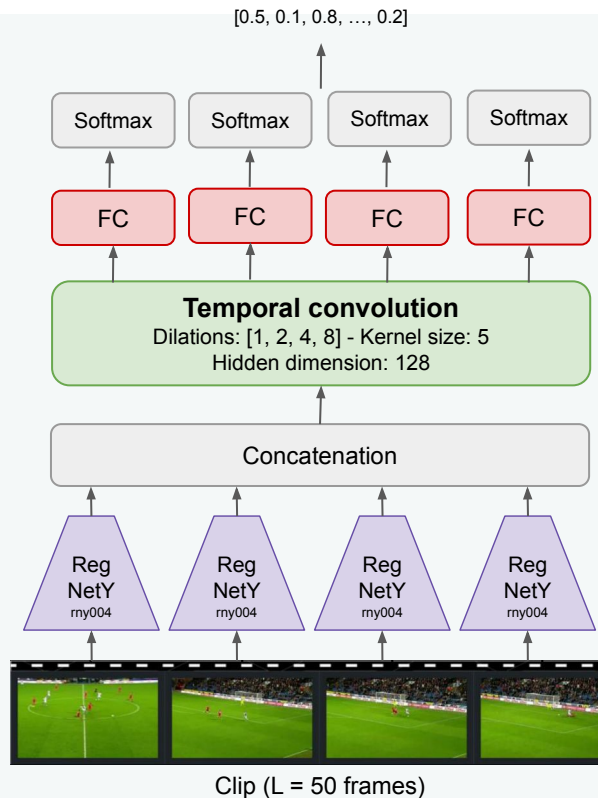
Unlike the U-Net ablations that suffered from pooling/striding information loss, the Dilated Temporal Convolution (1, 2, 4, 8 dilations) expands the receptive field without downsampling, maintaining the frame-perfect precision required for spotting.

Action Recognition over Frame Localization

The bottleneck in action spotting is confusing similar action classes. The TCN + Delta combination succeeds because it explicitly encodes a "motion signature" to differentiate classes, rather than relying on artificial label smoothing.

The Backbone

RegNetY-004 proved to be the sweet spot for spatial feature extraction, outperforming heavier backbones (like rynn008) while keeping parameters under 5M.



Parameters: **4.88M**

Data Aug.

- Horiz. Flips
- Gaussian blur
- Color Jitter

9. Quantitative Results

- Long temporal actions (such as cross) took the biggest hit
- Fast actions keep similar performances
- Rare classes are still not detected



Variant	AP10	AP12
Baseline	7.49	6.24
Best Week 6	36.02	35.82
Best Week 7	31.22	26.01

Class	AP Baseline	AP Week 6	AP Week 7
PASS	29.36	71.16	72.52
DRIVE	17.45	66.05	63.21
HEADER	5.07	46.53	40.47
HIGH PASS	12.01	63.37	59.07
OUT	1.3	9.35	7.21
CROSS	1.86	41.1	26.48
THROW IN	4.17	24.31	35.62
SHOT	2.87	25.14	27.13
BALL PLAYER BLOCK	0.78	13.18	3.63
PLAYER SUCCESSFUL TACKLE	0	0	0
FREE KICK	0	69.7	18.18
GOAL	0	0	0

10. Qualitative Results

Best Week 6



Best Week 7



11. Conclusions

Dataset was not ideal. Low quality, not homogeneous, unbalanced...



Temporal information is the key. Biggest gains.



AP-based model. More reliable than validation loss.



Future work: focus on stronger temporal context and dataset preprocessing

